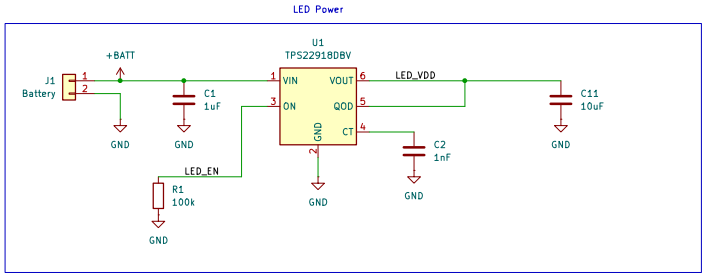
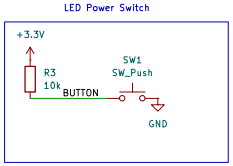
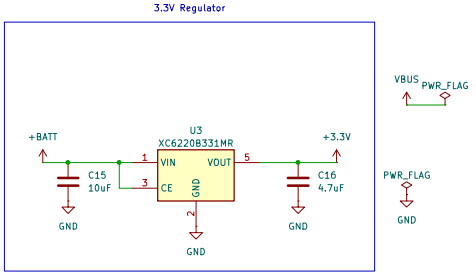
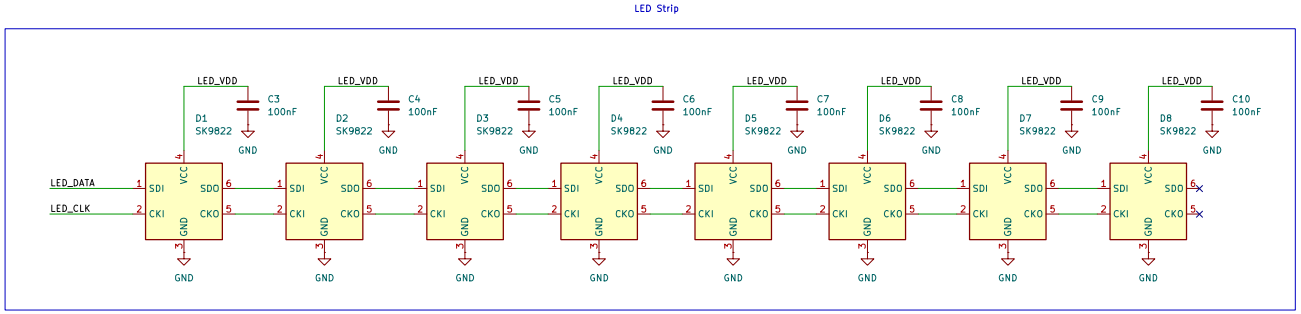
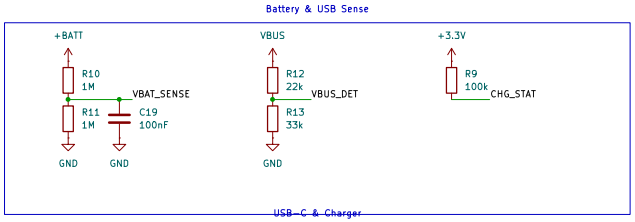
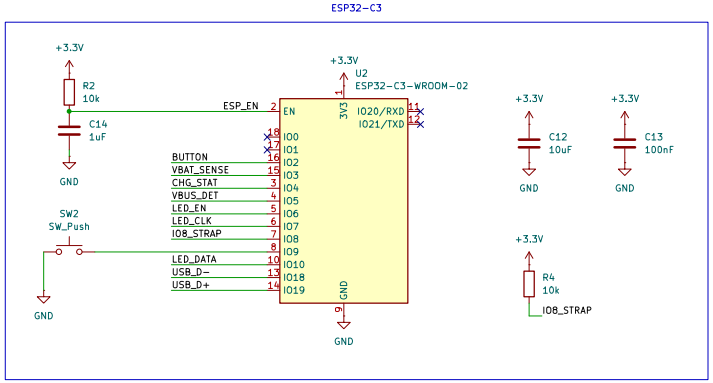

LED Power
J1
Battery
+BATT
GND
C1
1uF
U1
TPS22918DBV
VIN
ON
VOUT
QOD
CT
GND
LED_VDD
C2
1nF
C11
10uF
LED_EN
R1
100k

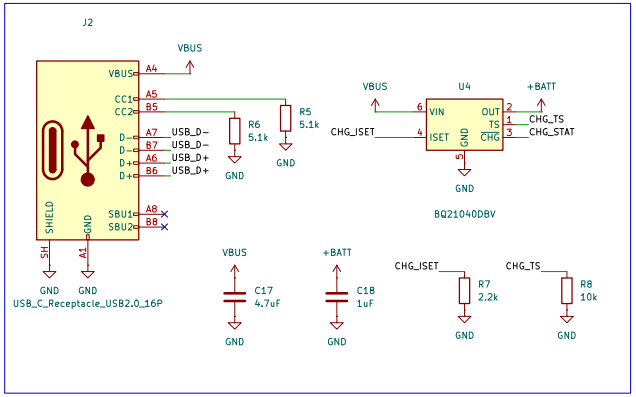

LED Power Switch
+3.3V
R3
10k
SW1
SW_Push
BUTTON
GND


3.3V Regulator
U3
XC6220B331MR
VIN
VOUT
CE
GND
+BATT
+3.3V
C15
10uF
C16
4.7uF
VBUS
PWR_FLAG


LED Strip
LED_VDD
D1
SK9822
D2
SK9822
D3
SK9822
D4
SK9822
D5
SK9822
D6
SK9822
D7
SK9822
D8
SK9822
C3
100nF
C4
100nF
C5
100nF
C6
100nF
C7
100nF
C8
100nF
C9
100nF
C10
100nF
LED_DATA
LED_CLK
SDI
SDO
CKI
CKO
VCC
GND


Battery & USB Sense
+BATT
R10
1M
R11
1M
C19
100nF
VBAT_SENSE
VBUS
R12
22k
R13
33k
VBUS_DET
+3.3V
R9
100k
CHG_STAT
GND
USB-C & Charger


ESP32-C3
+3.3V
R2
10k
C14
1uF
SW2
SW_Push
U2
ESP32-C3-WROOM-02
EN
3V3
IO0
IO1
IO2
IO3
IO4
IO5
IO6
IO7
IO8
IO9
IO10
IO18
IO19
IO20/RXD
IO21/TXD
GND
ESP_EN
BUTTON
VBAT_SENSE
CHG_STAT
VBUS_DET
LED_EN
LED_CLK
IO8_STRAP
LED_DATA
USB_D-
USB_D+
C12
10uF
C13
100nF
R4
10k


J2
VBUS
CC1
CC2
D-
D+
SBU1
SBU2
SHIELD
USB_C_Receptacle_USB2.0_16P
R5
5.1k
R6
5.1k
U4
BQ21040DBV
VIN
ISET
OUT
TS
CHG
CHG_ISET
CHG_TS
CHG_STAT
+BATT
C17
4.7uF
C18
1uF
R7
2.2k
R8
10k